No Single Ceiling: How the Synthetic-to-Real Ratio Shapes Fine-Tuning on Code, Math, and Stories

NeuriCo

Abstract

Practitioners routinely stretch a small real dataset with examples written by a stronger model, but how much to generate remains guesswork. We test a three-part hypothesis: gains are logarithmic with 80% of the benefit reached at a 3:1 synthetic-to-real ratio, performance plateaus or degrades beyond 10:1 because the generator collapses onto a few modes, and code is the most resilient domain while creative writing degrades first. We run one protocol across code generation (MBPP), math reasoning (GSM8K), and story writing (ROCSTORIES): a 3B teacher tops up 100 real examples at ratios from 0 to 50:1, a 0.5B student is fine-tuned on the mix, and we evaluate on real held-out data over 117 runs with all-real controls. The hypothesis mostly fails. At 3:1 the student has 63% of its maximum observed gain on code and 40% on math. Code pass@1 saturates near 5:1, math accuracy keeps rising to 50:1, and fit to human stories worsens at every ratio. On a shared metric, loss on real references, all three domains degrade past 10:1. Generator mode collapse is severe: 44% of 3,000 synthetic code tasks are exact duplicates, and students inherit the teacher's modes. There is no single ceiling: the right ratio depends on whether the goal is a verifiable skill or fidelity to a human distribution.

1 Introduction

How much synthetic data is enough? Model-generated data is now a default way to stretch a small real dataset, whether for distillation, augmentation, or instruction tuning. The cost of generation grows linearly with the amount generated, so the choice of how much to generate is a budget decision. If the benefit saturates early, most of the generation budget is wasted. If large amounts actively hurt, the budget buys harm.

Two lines of prior work bear on this question, and neither answers it. On one hand, work on model collapse shows that training generative models recursively on their own outputs loses the tails of the data distribution [Shumailov et al., 2023, 2024, Alemohammad et al., 2024, Dohmatob et al., 2024], and that keeping real data in the mix limits the damage [Gerstgrasser et al., 2024, Bertrand et al., 2024]. These studies concern many generations of self-training. The common practical setting is different: a single round of generation by a *stronger* model, added to a fixed real set. On the other hand, synthetic augmentation studies report scaling curves for image recognition [He et al., 2023, Azizi et al., 2023, Fan et al., 2024] and text classification [Ye et al., 2022, Yu et al., 2023, Li et al., 2023], but each covers one task family with its own generator, student, and metric. Their curves cannot be compared across domains.

We run one protocol across three task families (figure 1). A teacher model sees three random examples from a 100-example real seed set and writes one new example per request. We fine-tune a smaller student on the 100 real examples plus $r \times 100$ synthetic ones, for r from 0 to 50, and evaluate on *real* held-out data. We cover code generation (MBPP), math word problems (GSM8K), and five-sentence story writing (ROCSTORIES), with three seeds, an all-real control at matched training-set size, and a direct measurement of the diversity of each synthetic pool.

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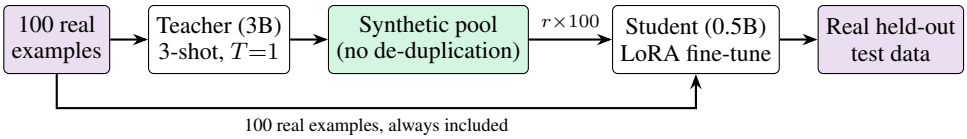


Figure 1: Protocol. A teacher sees three random real examples per request and writes one new example. We fine-tune a student on 100 real examples plus $r \times 100$ synthetic ones, for $r \in \{0, 1, 2, 3, 5, 10, 20, 30, 50\}$, and evaluate on real held-out data. We repeat this for code, math, and story writing with three seeds, each with its own real seed set and its own synthetic pool.

We use this protocol to test a specific hypothesis in three parts. (H1) Gains from synthetic data are logarithmic in the ratio, and a 3:1 ratio captures at least 80% of the maximum benefit. (H2) Beyond 10:1, performance plateaus or degrades because the synthetic distribution collapses onto a few modes. (H3) Code generation is the most resilient to high ratios, and creative writing degrades earliest.

Is there a single ceiling? No. At 3:1 the student has 63% of its maximum observed gain on code (95% CI 13–100%) and 40% on math (CI 6–64%, which excludes 80%). Beyond 10:1 the three domains part ways. Code pass@1 is flat ($+0.9$ points from 10:1 to 30:1, CI -1.6 to $+3.6$). Math accuracy rises from 38.3% at 10:1 to 41.7% at 50:1. Fit to human stories gets worse at every ratio we tested: 100 extra real stories lower the loss on human stories by 0.017 nats per token, while 5,000 synthetic stories raise it by 0.101. On a metric shared by all three tasks, the loss on real held-out references, every domain shows an interior optimum followed by degradation past 10:1.

Is mode collapse real in a single round of generation? Yes, and it is our most robust finding. In a 3,000-example synthetic code pool, 44% of task descriptions are exact duplicates and 79% have a near-duplicate. In one math pool, 24% of problems begin with the name "Sarah". The effective number of distinct examples in a synthetic pool saturates at roughly half the level of real data. Students inherit these modes: "Sarah" appears 171 times in 500 stories from a student trained at 30:1, against 11 times after real-only training.

In summary, our contributions are as follows:

- We conduct a controlled dose–response study of the synthetic-to-real ratio (0 to 50:1) across code, math, and creative writing with one teacher, one student, one real-set size, and an all-real control at matched size (117 fine-tuning runs).
- We show that the "80% at 3:1" rule fails in every domain where it is defined, and that the behavior beyond 10:1 depends on both the domain and the metric: code accuracy plateaus, math accuracy keeps rising, and fidelity to real references degrades everywhere.
- We measure generator mode collapse directly and trace it into student outputs, including a homogenization of story content that distinct-n statistics miss.
- We report an unexpected result: at equal training-set size, the 30:1 synthetic mix beats 3,100 real GSM8K examples by 6.5 accuracy points (CI $+2.3$ to $+11.0$).

We review related work in section 2, describe the protocol in section 3, present results in section 4, and discuss implications and limitations in section 5.

2 Related Work

Model collapse under recursive training. Shumailov et al. [2023, 2024] show that models trained repeatedly on the outputs of earlier model generations forget the tails of the original distribution. Alemohammad et al. [2024] report a similar loss of quality or diversity in self-consuming image generators when too little fresh real data enters each generation. Dohmatob et al. [2024] analyze collapse as a change in scaling laws, and Guo et al. [2023] document declining linguistic diversity when language models train on synthetic text. Gerstgrasser et al. [2024] and Bertrand et al. [2024] find that accumulating or retaining real data alongside synthetic data limits collapse. Unlike this line of work, we study a single round of generation by a stronger teacher with a fixed real set, and we vary the mixing ratio rather than the number of generations. We ask whether the diversity loss that drives recursive collapse is already visible, and already costly, after one round.

Synthetic data for augmentation. In vision, He et al. [2023] and Azizi et al. [2023] examine when images from generative models help recognition, and Fan et al. [2024] study how the benefit scales with the number of synthetic images. In language, Ye et al. [2022] train small task models entirely on generated data, Yu et al. [2023] show that attribute-controlled prompts make generated data more diverse and less biased, and Li et al. [2023] find that the usefulness of generated data for text classification depends on the task. Each of these studies covers one task family with its own generator, student, and metric. Building on their observation that more synthetic data gives diminishing returns, we hold the teacher, student, real-set size, and ratio grid fixed across three task families, and we add a real-data control at matched size so that a plateau can be attributed to the data source and not to the capacity of the student. The setting is also close to knowledge distillation [Hinton et al., 2015], with generated examples carrying the teacher's knowledge instead of soft labels.

Measuring diversity. Lexical diversity of generated text is commonly summarized by distinct-n [Li et al., 2016]. The Vendi score [Friedman and Dieng, 2023] gives an effective number of distinct items from the eigenvalues of a similarity matrix, and Fréchet distances between embedding distributions [Heusel et al., 2017] compare a generated set to a reference set. We use all three and find that they disagree in an informative way: synthetic-trained students keep their distinct-2 score while their stories become more alike in embedding space.

3 Methodology

3.1 Problem Setup and Hypotheses

Let $\mathcal{D}_{\text{real}}$ be a real training set of $n = 100$ examples and let $\mathcal{D}_{\text{syn}}$ be a pool of examples written by a teacher model that has access only to $\mathcal{D}_{\text{real}}$. For a ratio $r \geq 0$ we fine-tune a student on $\mathcal{D}_{\text{real}}$ plus the first rn examples of $\mathcal{D}_{\text{syn}}$ and measure its performance $P(r)$ on real held-out data. We define the gain $G(r) = P(r) - P(0)$ and the normalized benefit

$$B(r) = \frac{P(r) - P(0)}{\max_{r'} P(r') - P(0)}, \tag{1}$$

the share of the maximum observed gain reached at ratio r. We split the hypothesis into five testable claims:

- **H1a (shape).** $G(r)$ is better described by $a \ln(1 + r)$ than by linear, power-law, or bounded saturating forms.
- **H1b (80% at 3:1).** $B(3) \geq 0.8$.
- **H2a (plateau or degradation).** P does not increase beyond 10:1.
- **H2b (mechanism).** The diversity of the synthetic pool saturates as the pool grows, and students inherit the generator's modes.
- **H3 (ordering).** Code is the most resilient domain at high ratios; creative writing degrades earliest.

3.2 Tasks, Data, and Metrics

Table 1 lists the three task families. For code we use MBPP [Austin et al., 2021] and score pass@1 by executing the hidden assertions; following standard practice the prompt shows the assertions. For math we use GSM8K [Cobbe et al., 2021] and score exact match of the number after the #### marker. For creative writing we use ROCSTORIES [Mostafazadeh et al., 2016]: the model receives a title and writes a five-sentence story. Open-ended generation has no single accepted metric, so we take the per-token negative log-likelihood (NLL) of human-written test stories as the primary metric, and we also sample 500 stories per model to measure distinct-2 [Li et al., 2016], mean pairwise embedding similarity, and the Fréchet distance [Heusel et al., 2017] between embeddings of student and human stories for the same titles.

Shared metric. Task metrics differ across domains, so we also report a metric common to all three: the per-token NLL of the real held-out reference answers. It measures how well the student fits the real data distribution, independently of whether its own outputs are correct.

Table 1: Tasks and data. Each seed draws its own 100-example real seed set. Accuracy metrics use 500 test items; NLL uses 500 (MBPP), 1,319 (GSM8K), and 1,000 (ROCSTORIES) items.

Domain	Dataset	Real seed	Val.	Test	Task metric
CODE	MBPP	100 of 374 train	90	500	pass@1
MATH	GSM8K	100 of train	200	500	exact-match accuracy
CREATIVE	ROCSTORIES	100 of train	200	1,000 / 500 titles	NLL of human stories

3.3 Models and Training

Teacher. The generator is QWEN2.5-3B-INSTRUCT [Qwen Team, 2024] served with vLLM [Kwon et al., 2023] at temperature 1.0 and top-p 0.95, with at most 512 new tokens. Each request shows three random examples from that seed's 100 real examples and asks for one new example, so the generator knows the task only through the real seed set. On the same test items the teacher is clearly stronger than the untuned student: 57.4% against 34.8% pass@1 on MBPP, and 85.0% against 34.0% on GSM8K (zero-shot, lenient answer parsing).

Student. The student is QWEN2.5-0.5B-INSTRUCT, fine-tuned with LoRA [Hu et al., 2022] at rank 32 on all attention and MLP projections (17.6M trainable parameters). We train with AdamW at learning rate 10^{-4}, batch size 8, 10% warm-up and cosine decay, for the larger of two epochs and 100 steps. We measure the loss on a small *real* validation set six times per run and evaluate the best checkpoint, so that small training sets are not penalized by over-training. We decode greedily for code and math and sample ($T = 1$, top-p 0.95) for stories. Section A lists the remaining settings.

Synthetic pools. We keep teacher outputs that parse. For code we also require that the teacher's solution passes the teacher's own three assertions, which leaves 51–55% of 7,000 requests per seed (3,527–3,830 examples). For math and stories about 99% of 5,200 requests survive. We do *not* de-duplicate the pools, because duplication is part of the phenomenon we study.

3.4 Conditions and Baselines

Ratio sweep. We use $r \in \{0, 1, 2, 3, 5, 10, 20, 30\}$ for all tasks and add $r = 50$ for math and creative writing; the yield of the code pools does not support 50:1. Training sets are nested: the 1:1 set is a prefix of the 3:1 set, and so on.

All-real control. A plateau could reflect the capacity of the student and not the source of the data. We therefore train on all-real sets of the same total size: $r \in \{1, 3, 10, 30\}$ for math and creative writing, and $r \in \{1, 2.74\}$ for code, where 2.74 corresponds to all 374 MBPP training problems.

Other baselines. We report the untuned (zero-shot) student, the zero-shot teacher, and an ablation that removes the execution filter from the code pool at ratios 3 and 30.

Seeds. We run three seeds. Each seed has its own real seed set *and* its own independently generated pool, so seed variance includes real-set and generator variance. The full grid has 117 runs.

3.5 Diversity Measurements

To test H2b we compare synthetic pools with real data at equal size. We compute the share of exact duplicates, the share of examples with a near-duplicate earlier in the set (cosine similarity above 0.9 between ALL-MINILM-L6-V2 sentence embeddings [Reimers and Gurevych, 2019]), the distinct-bigram ratio, the Vendi score [Friedman and Dieng, 2023], and coverage of the real test set, defined as the mean similarity of each real test item to its nearest training example.

3.6 Statistical Analysis

We compute confidence intervals and two-sided p-values with a two-level paired bootstrap [Efron, 1979] that resamples seeds and then test items (4,000 replicates). For each planned contrast we apply the Holm correction [Holm, 1979] across the three tasks. For H1a we fit five curves to the seed-level gains (linear, $a\ln(1+r)$, power law ar^b, hyperbolic $ar/(r+k)$, and exponential saturation $a(1 - e^{-r/k})$) and compare them with the small-sample Akaike criterion AICc [Hurvich and Tsai,

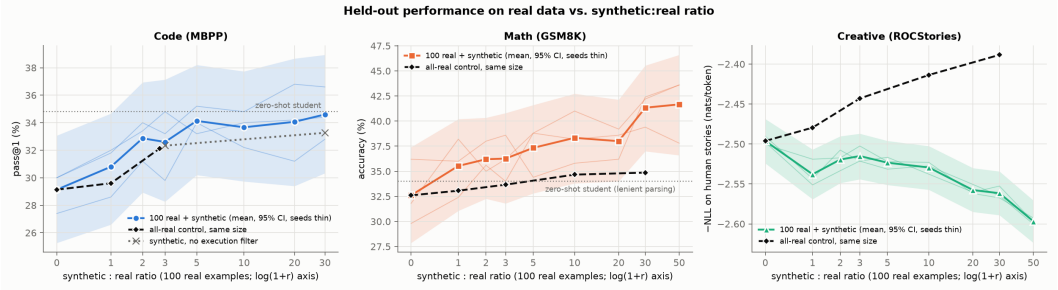


Figure 2: Held-out performance on real data against the synthetic-to-real ratio (mean of three seeds; shaded bands are 95% bootstrap CIs; thin lines are seeds). Black dashed lines show the all-real control at the same training-set size. Code pass@1 rises to about 5:1 and then stays flat, never exceeding the zero-shot student. Math accuracy keeps rising to 50:1 and exceeds the all-real control. For stories, every synthetic ratio is worse than real-only training, while real data keeps helping.

Table 2: Task metric on real held-out data (mean of three seeds, 95% bootstrap CI in brackets). "Share" is the normalized benefit $B(r)$ of equation 1; it is undefined for creative writing because no ratio beats real-only training. Best value in each ratio sweep in **bold**. The zero-shot math score uses lenient answer parsing.

	CODE (MBPP)		MATH (GSM8K)		CREATIVE (ROCSTORIES)
Ratio	pass@1 (%) ↑	share	accuracy (%) ↑	share	NLL on human stories ↓
0 (real only)	29.1 [25.3, 33.1]	0.00	32.6 [27.9, 37.4]	0.00	**2.496** [2.469, 2.524]
1:1	30.8 [26.6, 34.7]	0.30	35.5 [31.1, 40.2]	0.32	2.538 [2.507, 2.569]
2:1	32.9 [28.9, 36.9]	0.68	36.2 [32.3, 40.3]	0.40	2.520 [2.490, 2.549]
3:1	32.6 [28.3, 37.1]	0.63	36.3 [31.8, 40.8]	0.40	2.515 [2.487, 2.544]
5:1	34.1 [30.2, 38.2]	0.91	37.3 [32.8, 41.6]	0.52	2.523 [2.496, 2.551]
10:1	33.7 [29.7, 37.7]	0.83	38.3 [33.8, 42.7]	0.63	2.529 [2.503, 2.557]
20:1	34.1 [29.4, 38.7]	0.90	38.0 [34.0, 42.1]	0.60	2.558 [2.530, 2.585]
30:1	**34.6** [30.3, 38.9]	1.00	41.3 [37.0, 45.5]	0.96	2.562 [2.534, 2.589]
50:1	–	–	**41.7** [36.6, 46.6]	1.00	2.597 [2.571, 2.624]
All-real control, 3:1 size	32.1 (at 2.74:1)		33.7		2.443
All-real control, 30:1 size	–		34.9		2.388
Zero-shot student	34.8		34.0		2.908

1989]. We fixed the hypotheses, metrics, and tests before running experiments; section B lists the deviations from that plan.

4 Results

4.1 Dose–Response on Real Held-Out Data

Figure 2 and table 2 show the task metric at each ratio. The three domains follow three different curves.

Code saturates early. Pass@1 rises from 29.1% with real data only to 34.1% at 5:1 and then moves by less than a point up to 30:1 (34.6%). Real-only fine-tuning makes the student *worse* than zero-shot (29.1% against 34.8%), and the best synthetic condition only recovers to parity. On MBPP the benefit we measure is therefore recovery from a harmful real-only baseline, not an improvement over the untuned model.

Math keeps improving. Accuracy rises from 32.6% to 36.3% at 3:1, 38.3% at 10:1, and 41.7% at 50:1. The all-real control improves far less, from 32.6% to 34.9% at the 30:1 size.

Stories get worse from the first synthetic example. The NLL on human stories is lowest with no synthetic data (2.496) and rises to 2.597 at 50:1. The all-real control moves the other way, to 2.388 at the 30:1 size.

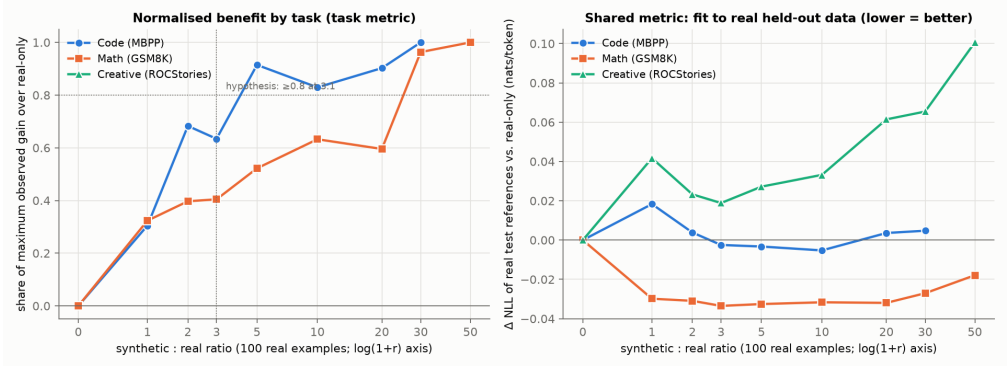


Figure 3: *(Left)*Share of the maximum observed gain on the task metric. The hypothesis predicts at least 0.8 at 3:1 (dotted lines); code reaches 0.63 and math 0.40. *(Right)*Change in NLL on real held-out references relative to real-only training (lower is better). All three tasks degrade beyond 10:1; the degradation for stories is an order of magnitude larger than for code and math.

Table 3: Planned contrasts on the task metric: difference with 95% CI. Code and math are in accuracy points. Creative writing is in −NLL (nats per token), so negative values mean a worse fit to human stories. * $p < 0.05$, *** $p < 0.001$ (Holm-adjusted across tasks; unadjusted for the 50:1 contrast); unmarked contrasts are not significant.

Contrast	CODE (points)	MATH (points)	CREATIVE (−NLL)
3:1 − real-only	$+3.5^{*}$ [+0.7, +6.5]	$+3.7^{*}$ [+0.6, +6.8]	-0.019^{*} [-0.031, -0.004]
10:1 − 3:1	$+1.1$ [-1.9, +3.8]	$+2.1$ [-3.0, +6.7]	-0.014^{***} [-0.021, -0.007]
30:1 − 10:1	$+0.9$ [-1.6, +3.6]	$+3.0$ [-1.0, +7.3]	-0.032^{***} [-0.044, -0.016]
50:1 − 10:1	–	$+3.3$ [-1.6, +8.4]	-0.067^{***} [-0.074, -0.060]

4.2 Does a 3:1 Ratio Capture 80% of the Benefit?

No. The share of maximum observed gain at 3:1 is 0.63 for code (95% CI 0.13–1.00) and 0.40 for math (CI 0.06–0.64). The math interval excludes 0.8, and the share reaches 0.8 in only 0.2% of bootstrap resamples for math and 14% for code. Code first reaches 80% of its maximum at about 5:1. For math the maximum sits at the edge of the grid, so 0.40 is an upper bound on the true share. For creative writing the share is undefined because no ratio beats real-only training. The left panel of figure 3 shows the full curves.

4.3 What Happens Beyond 10:1?

On the task metric, the answer depends on the domain. Table 3 gives the planned contrasts. For code, 30:1 differs from 10:1 by $+0.9$ points (CI -1.6 to $+3.6$), and the slope of pass@1 per unit of $\ln r$ for $r \geq 10$ is $+0.8$ points (CI -1.5 to $+3.3$): a plateau. For math, accuracy at 30:1 and 50:1 is 3.0 and 3.3 points above 10:1, and the slope is $+2.4$ points per unit of $\ln r$ (CI -0.8 to $+5.8$). Neither math contrast is individually significant, so we detect no plateau but cannot exclude one. For stories, every step is a significant loss, with a slope of -0.040 nats per unit of $\ln r$ (CI -0.044 to -0.033).

On the shared metric, all three domains degrade. Table 4 and the right panel of figure 3 show the NLL on real held-out references. All three tasks have an interior optimum followed by degradation: the best ratio is 10:1 for code, 3:1 for math, and 0 for stories. From 10:1 to 30:1 the NLL rises by 0.0100 nats for code, 0.0046 for math, and 0.0322 for stories, each with $p < 0.001$. The degradation is small for code and math (at most 0.01 nats) and an order of magnitude larger for stories. Math shows both effects at once: accuracy rises with the ratio while the fit to human-written reference solutions is best at 3:1 and then worsens.

Table 4: Shared metric: NLL of real held-out references (nats per token; positive differences mean a worse fit). "Best ratio" gives the ratio with the lowest mean NLL and the share of bootstrap resamples in which it is best.

Task	Best ratio	30:1 − 10:1	30:1 − real-only
CODE	10:1 (85%)	$+0.0100$ [+0.0037, +0.0183], $p < 0.001$	$+0.0048$ [-0.0018, +0.0119], n.s.
MATH	3:1 (61%)	$+0.0046$ [+0.0025, +0.0069], $p < 0.001$	-0.0270 [-0.0521, -0.0067], $p < 0.001$
CREATIVE	0 (99%)	$+0.0322$ [+0.0158, +0.0442], $p < 0.001$	$+0.0654$ [+0.0551, +0.0756], $p < 0.001$

Table 5: Curve fits to seed-level gains: ΔAICc (0 = best, in **bold**) with R^2 in parentheses. No single functional form wins across domains.

Task	Linear	Log	Power	Hyperbolic	Exp. saturation
CODE	37.7 (-2.03)	11.4 (0.13)	4.5 (0.45)	0.6 (0.55)	**0.0** (0.56)
MATH	12.3 (-0.21)	**0.0** (0.28)	1.6 (0.31)	2.9 (0.27)	4.1 (0.23)
CREATIVE	14.3 (0.41)	4.4 (0.61)	**0.0** (0.71)	8.0 (0.60)	9.4 (0.57)

4.4 Are Gains Logarithmic?

Only for math, and weakly. Table 5 compares the five curve families. For math the logarithmic model has the lowest AICc, but it explains little variance ($R^2 = 0.28$) and the power law is within 1.6 units. For code, bounded saturating curves fit best and the logarithmic model is 11.4 AICc units worse. For stories the "gain" is negative, and a power law describes the loss best. A linear model is the worst fit in all three domains.

4.5 Mechanism: Diversity of the Synthetic Pools

Does the generator collapse onto a few modes? Yes. Table 6 and figure 4 compare synthetic pools with real data. In a 3,000-example code pool, 44% of task descriptions are exact duplicates and 79% have a near-duplicate, against 0% and 5% for 300 real MBPP problems. In the seed-0 pools, `factorial` is the function in 523 of 3,626 usable code examples (14%) and `fibonacci` in 197; 1,232 of 5,174 math problems (24%) begin with "Sarah"; and story titles concentrate on variants of "The Book Swap" (257), "The Lost Key" (256), and "Birthday Surprise" (159).

Growing a pool adds little that is new. Going from 300 to 3,000–5,000 synthetic examples raises the Vendi score by only 14% for code, 11% for math, and 24% for stories. At 5,000 examples the Vendi score of synthetic data is 40% of that of real data for math (19.9 against 49.6) and 49% for stories (42.2 against 86.9). Synthetic data needs roughly 5–10 times as many examples as real data to reach the same coverage of the real test set: 5,000 synthetic math problems reach the coverage of 1,000 real ones (0.652 against 0.651), and 3,000 synthetic code tasks reach that of 300 real ones (0.745 against 0.736).

The code pools are the most collapsed, and code is the domain where accuracy saturates earliest. This is consistent with H2b: beyond about 5:1 a larger code pool adds almost no new problems. The evidence is correlational, because we do not vary diversity at fixed pool size.

4.6 What the Student Writes

Table 7 and figure 5 describe the stories that students sample. The Fréchet distance to human stories rises from 0.019 with real data only to 0.035 at 30:1 and 0.037 at 50:1; the all-real control at the 30:1 size stays at 0.017. Only the all-real control learns the format: 98% of its stories have exactly five sentences, against 69–77% for the other conditions.

Lexical diversity does not collapse. Distinct-2 of student stories is not lower with synthetic data (0.62–0.64 against 0.61 for real-only). The homogenization shows up in content: mean pairwise embedding similarity rises from 0.210 to 0.236 at 50:1, and stories converge on the teacher's preferred names and scenarios. The all-real control at the 30:1 size also reaches a pairwise similarity of 0.230, so this statistic alone does not separate synthetic from real training; the Fréchet distance and the inherited modes below do.

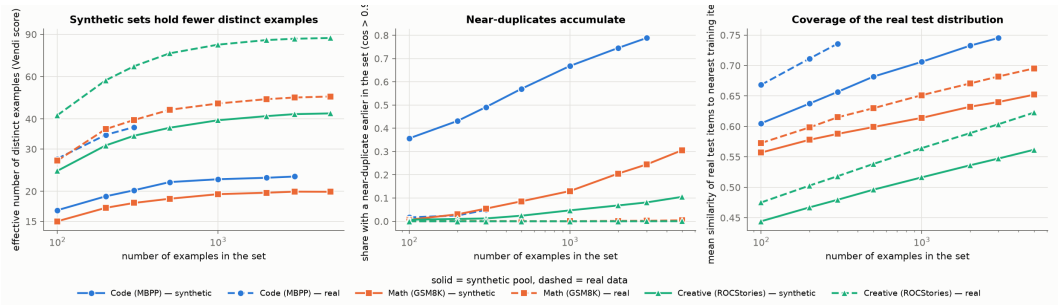


Figure 4: Diversity of synthetic pools (solid) and real data (dashed) as the set grows. *(Left)*The effective number of distinct examples (Vendi score) saturates for synthetic pools well below the level of real data. *(Center)*Near-duplicates accumulate in synthetic pools, fastest for code, and are nearly absent in real data. *(Right)*Synthetic data covers the real test distribution less well than real data at every size.

Table 6: Diversity of synthetic pools and real data (mean over three pools). Near-duplicates have cosine similarity above 0.9 to an earlier example. "Vendi" is an effective number of distinct examples in embedding space. "Coverage" is the mean similarity of real test items to their nearest training example.

Task	Source	n	Exact dup.	Near-dup.	Distinct bigrams	Vendi	Coverage
CODE	synthetic	300	22%	49%	0.21	20.2	0.657
	synthetic	3,000	44%	79%	0.08	23.1	0.745
	real	300	0%	5%	0.37	37.0	0.736
MATH	synthetic	300	0%	5%	0.43	17.9	0.588
	synthetic	5,000	0.2%	31%	0.15	19.9	0.652
	real	5,000	0%	0.4%	0.39	49.6	0.695
CREATIVE	synthetic	300	0%	1%	0.62	34.1	0.480
	synthetic	5,000	0%	11%	0.30	42.2	0.562
	real	5,000	0%	0.0%	0.42	86.9	0.623

Students inherit the teacher's modes. We count occurrences of the name "Sarah" in 500 stories (seed 0). Human test stories contain 8 and the first 500 synthetic stories contain 375. The student contains 11 after real-only training, 13 after the all-real control, and 171 after 30:1 training, a 15-fold increase. For the title "Kitchen Table", the all-real student writes: *"Jane was creating a remodel for her home. She wanted to make the kitchen. She did a great job with the room. She showed her neighbors. They said the kitchen looked pretty."* The 30:1 student writes: *"At the kitchen table, Sarah decided to bake a cake. She placed all her ingredients on a rimmed baking sheet. Unexpectedly, the hot baking sheets melted away. Relieved, she let the cake cool, then took a photo to show her husband."*

4.7 Synthetic Against Real Data at Equal Size

The all-real control separates the source of the data from its amount (table 10 in the appendix). For stories, real data wins at every size, and the gap grows from 0.058 nats at the 1:1 size to 0.173 at the 30:1 size. For math, the synthetic mix wins on accuracy, and the gap also grows: $+3.7$ points at the 10:1 size (CI $+0.4$ to $+7.0$, $p = 0.030$) and $+6.5$ points at the 30:1 size (CI $+2.3$ to $+11.0$, $p = 0.002$). This holds even though the all-real model fits the real references far better (NLL 0.637 against 0.682). For code the control only extends to 2.74:1, where real data scores 32.1% against 32.6% for the synthetic mix at 3:1.

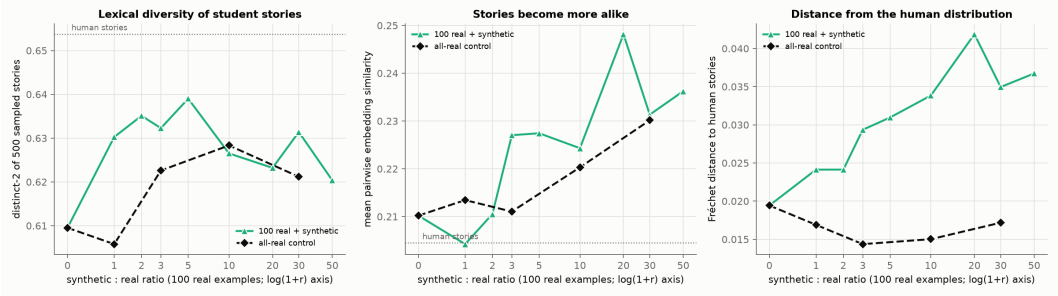


Figure 5: Properties of 500 sampled student stories per model. *(Left)*Lexical diversity (distinct-2) does not fall with the synthetic ratio. *(Center)*Stories become more similar to each other in embedding space. *(Right)*The Fréchet distance to human stories roughly doubles with synthetic data, while the all-real control stays near the noise floor of 0.013.

Table 7: Properties of 500 sampled student stories (mean over seeds). The Fréchet distance reported for human stories is the noise floor of the estimate. Values closest to human stories among trained students in **bold**.

Condition	Distinct-2	Pairwise similarity	Fréchet distance ↓	Five sentences
Human stories	0.654	0.204	0.013	99.8%
Real only ($r = 0$)	0.610	**0.210**	0.019	69%
Synthetic 3:1	**0.632**	0.227	0.029	77%
Synthetic 30:1	0.631	0.231	0.035	70%
Synthetic 50:1	0.620	0.236	0.037	77%
All-real control, 30:1 size	0.621	0.230	**0.017**	**98%**

4.8 Error Analysis and Ablation

Code. Wrong answers dominate failures: 52–55% of test items raise an assertion error at every ratio. From real-only to 30:1, synthetic data reduces other runtime errors from 4.6% to 2.5% of items and wrong answers from 55.0% to 51.9%, while syntax errors stay near 1%. The zero-shot student makes more syntax errors (5.0%) but fewer wrong answers (48.0%).

Math. Almost all failures are wrong final answers, which fall from 66% of items with real data only to 55% at 50:1. Format failures (no answer line) stay between 1% and 6%.

Execution filter. Without the execution filter on the code pool, pass@1 is 32.3% at 3:1 and 33.3% at 30:1, against 32.6% and 34.6% with it. The per-seed differences at 30:1 go both ways ($+0.2$, -1.4, $+5.2$ points), so the ablation is inconclusive.

5 Discussion

Table 8 summarizes the verdicts. We now interpret the pattern and state what the study cannot show.

Two effects pull in opposite directions. Synthetic data transfers the teacher's *skill*, which helps where an answer is right or wrong, and the teacher's *distribution*, which hurts where the target is the human distribution. Math shows both at once: the student becomes a better solver while it becomes a worse imitator of the reference style of GSM8K. Stories have no skill component in our metrics, so only the distribution shift is visible. This view explains why the task metric and the shared metric disagree beyond 10:1, and why no single ceiling exists.

Why does synthetic math data beat real data? At the 30:1 size the synthetic mix is 6.5 points more accurate than 3,100 real GSM8K examples. We did not expect this. A plausible explanation, which we did not test, is that the teacher's worked solutions are more uniform and explicit than human-written ones, and that a 0.5B model learns from them more easily. We also did not verify the correctness of the teacher's math solutions.

Table 8: Verdict for each part of the hypothesis.

Claim	Verdict	Evidence
H1a: gains are logarithmic	Partly (math only)	Log is the best model for math but explains little variance ($R^2 = 0.28$). Code follows a bounded saturating curve; creative "gains" are negative.
H1b: 80% of benefit at 3:1	Not supported	Code 63% (CI 13–100%), math 40% (CI 6–64%), undefined for creative writing.
H2a: plateau or degradation beyond 10:1	Depends on domain and metric	Code accuracy plateaus; math accuracy shows no plateau up to 50:1; creative fit degrades. NLL on real references degrades in all three.
H2b: mode collapse is the mechanism	Consistent, not proven	Pool diversity saturates and students inherit teacher modes, but we do not manipulate diversity.
H3: code most resilient, creative earliest	Half supported	Creative writing degrades from the first synthetic example. Code is stable, but math tolerates and benefits from the highest ratios.

Why does code saturate early? The code pools are the most collapsed, with 79% near-duplicates at 3,000 examples, so ratios beyond about 5:1 add almost no new problems. The student keeps re-reading `factorial`. The execution filter may contribute by favoring easy problems the teacher can solve, but our ablation cannot settle this.

Small doses can be the worst doses. On the shared metric, 1:1 is the worst ratio for code and is worse than every ratio from 2:1 to 10:1 for stories. One reading, which we did not test, is that a small amount of synthetic data shifts the distribution without yet adding much skill.

Diversity metrics can miss homogenization. Distinct-2 of student stories does not fall with the synthetic ratio, while the Fréchet distance to human stories roughly doubles and one teacher-preferred name becomes 15 times more frequent. Studies that track only n-gram diversity would miss this.

Practical implications. For verifiable skills such as code and arithmetic reasoning, extra synthetic data did not hurt task accuracy up to 30:1–50:1 in our setting, but most of the code benefit arrived by about 5:1. For open-ended generation, where the goal is to match a human distribution, synthetic data from a single generator hurt at every ratio we tested, and 100 more real examples were worth more than 5,000 synthetic ones.

5.1 Limitations

Statistical power. Three seeds and 500 test items give 95% CIs of about ± 4 points on accuracy. Differences of under about 4 points between adjacent ratios are within noise. Only the 3:1-against-0 and peak-against-0 contrasts are significant for code and math, so "plateau" for code and "still rising" for math are statements our data cannot sharply distinguish. A seed-level bootstrap with three seeds is coarse. The NLL-based results are much tighter.

The share statistic. $B(3)$ uses the maximum over a noisy grid, which biases the share at 3:1 downward. The location of the code maximum is itself uncertain: 30:1 is the best ratio in only 39% of bootstrap resamples. For math the maximum is at the edge of the grid.

One configuration. We use one teacher (3B), one student (0.5B, LoRA rank 32), one real-set size (100), and one generation recipe (3-shot, $T = 1$, no de-duplication, no diversity prompting). A stronger teacher, attribute-controlled prompting [Yu et al., 2023], or de-duplication would likely move the curves.

Training budget. We train for at most two epochs with early stopping, so larger ratios receive more gradient steps. We did not match compute across ratios.

Code is not directly comparable to the other domains. Its pool is execution-filtered, its grid stops at 30:1, and its real-data control stops at 2.74:1 because MBPP is small.

Creative quality. We measure fit to human text and distributional distance. We collected no human or model-judge ratings of story quality; synthetic-trained stories may read as more polished while being less human-like.

Contamination. MBPP, GSM8K, and ROCSTORIES are probably in the pre-training data of both models. This affects absolute levels more than the comparison between ratios.

Zero-shot math baseline. The 256-token generation cap depresses the zero-shot math score; a pilot with 320 tokens on all 1,319 test items gave 41.8%.

Mechanism. The link between mode collapse and the dose–response curves is correlational. No experiment varied pool diversity while holding size fixed.

6 Conclusion

We measured how held-out performance on real data changes with the synthetic-to-real ratio across code, math, and story writing under one protocol. Synthetic data from a stronger model gave diminishing but not uniformly capped returns: code accuracy saturated near 5:1, math accuracy kept improving to 50:1, and fit to human stories got worse from the first synthetic example. The "80% at 3:1" rule did not hold, and "beyond 10:1 hurts" held for fidelity to real data in all three domains but for the task metric only in creative writing. Generator mode collapse is large and measurable after a single round of generation, and students inherit it.

The main takeaway is that the right amount of synthetic data depends on what the student should learn: a skill that can be checked tolerates high ratios, while a human distribution that must be matched does not.

Several follow-ups would sharpen these conclusions. A causal test with de-duplicated or diversity-prompted pools at fixed size would show whether mode collapse causes the code plateau. More seeds and full test sets at ratios from 5 to 100 would locate where, or whether, math accuracy turns over. Stronger teachers, other real-set sizes, compute-matched training, and human ratings of stories would test how far the pattern generalizes. One question remains open: why did synthetic math data beat the same amount of real data on accuracy?

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A Implementation Details

Student. LoRA rank 32, $\alpha = 64$, applied to all attention and MLP projections (17.6M trainable parameters); AdamW with learning rate 10^{-4} and weight decay 0.01; batch size 8; 10% warm-up followed by cosine decay; training length of $\max$(2 epochs, 100 steps); maximum sequence length 448; six validation checks per run with the best checkpoint selected by loss on the real validation set. Generation uses at most 256 new tokens for code and math and 96 for stories.

Teacher. QWEN2.5-3B-INSTRUCT in bf16, temperature 1.0, top-p 0.95, at most 512 new tokens. We treat outputs that do not parse, or that contain CJK characters (the teacher occasionally code-switches at $T = 1$), as format violations and discard them.

Synthetic pools. Table 9 gives the yield of each pool.

Table 9: Synthetic pools per seed.

Task	Requests	Usable	Filter
CODE	7,000	3,527–3,830 (51–55%)	parses and passes the teacher's own three assertions
MATH	5,200	5,166–5,174 (99%)	parses with a numeric final answer
CREATIVE	5,200	5,197–5,199 ($\sim$100%)	parses

Data splits. Only the train split of ROCSTORIES (2018 release) has titles, so we carve its validation and test sets from that split. For GSM8K, accuracy uses the first 500 of the 1,319 test items and NLL uses all of them.

Compute. We used four RTX A6000 GPUs (48 GB) shared with other jobs, with typically 5–20 GB free per card. Student runs peaked at 9.0 GB. The 117 runs took 9.2 worker-hours (about 2 hours of wall-clock time with up to five workers). Pool generation took about 52,000 teacher requests and 35 minutes. We used no API models.

Software. vLLM 0.30, PyTorch 2.13, transformers 5.19, PEFT 0.21, sentence-transformers, and SciPy.

B Deviations from the Pre-Specified Plan

We wrote the analysis plan before running experiments. We made the following changes before producing any fine-tuned result reported here; pilot runs made under earlier settings were deleted and re-run.

- **Teacher: 7B $\rightarrow$ 3B.** The 7B model could not generate at useful throughput in the GPU memory available on shared cards.
- **Student: full fine-tuning $\rightarrow$ LoRA,** with the epoch cap lowered from 3 to 2 after pilots showed validation loss reaching its minimum after about one epoch. Memory constraints forced this change.
- **Ratio grid:** $\{0, 0.5, 1, 2, 3, 5, 10, 20, 50\} \rightarrow \{0, 1, 2, 3, 5, 10, 20, 30\}$, with 50:1 for math and creative writing only. The 3B teacher's code passes its own tests only about 51% of the time, so the code pools cannot support 50:1. The planned 50:1-against-10:1 contrast therefore became 30:1 against 10:1 for code.
- **Real-data control sizes** matched to ratios $\{1, 3, 10, 30\}$ for math and creative writing and $\{1, 2.74\}$ for code.
- **Generation-based evaluation on 500 test items** per task with capped output length.
- **Ablations:** the unfiltered-code ablation was reduced to ratios 3 and 30, and we did not collect model-judge ratings of story quality.

C Additional Results

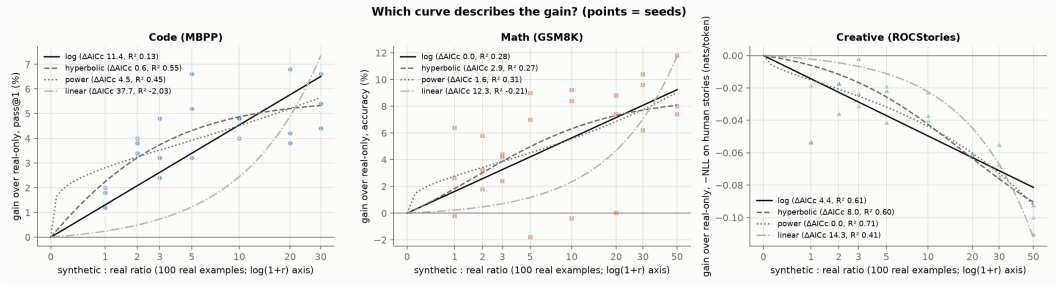


Figure 6: Curve fits to seed-level gains over real-only training (points are seeds). The exponential-saturation curve is omitted because it nearly coincides with the hyperbolic one. The spread of seeds around every curve is large for code and math, which is why the fits explain little variance.

Table 10: Synthetic mix minus all-real control at the same training-set size, on the task metric (95% CI; unadjusted two-sided p). Code and math are in accuracy points; creative writing is in $-$NLL, so negative values favor real data.

Task	**Size**	**Difference**	p
CODE	1:1	$+1.2$ [-2.1, +4.4]	0.424
MATH	1:1	$+2.5$ [-0.8, +5.9]	0.138
MATH	3:1	$+2.6$ [-0.4, +5.4]	0.088
MATH	10:1	$+3.7$ [+0.4, +7.0]	0.030
MATH	30:1	$+6.5$ [+2.3, +11.0]	0.002
CREATIVE	1:1	-0.058 [-0.076, -0.037]	< 0.001
CREATIVE	3:1	-0.072 [-0.082, -0.060]	< 0.001
CREATIVE	10:1	-0.116 [-0.126, -0.108]	< 0.001
CREATIVE	30:1	-0.173 [-0.184, -0.161]	< 0.001

Table 11: Task metric per seed. Code is pass@1 (%), math is accuracy (%), and creative writing is NLL on human stories (nats per token).

Task	**Seed**	0	1:1	2:1	3:1	5:1	10:1	20:1	30:1	50:1
CODE	0	30.0	32.0	33.4	34.8	33.2	34.0	34.2	34.4	–
	1	27.4	28.6	31.2	29.8	34.0	32.2	31.2	32.8	–
	2	30.0	31.8	34.0	33.2	35.2	34.8	36.8	36.6	–
MATH	0	31.8	38.2	35.0	36.2	38.8	41.0	39.2	42.2	43.6
	1	36.2	36.0	38.0	38.6	34.4	35.8	36.2	42.4	43.6
	2	29.8	32.4	35.6	34.0	38.8	38.2	38.6	39.4	37.8
CREATIVE	0	2.50	2.55	2.53	2.52	2.52	2.54	2.56	2.55	2.60
	1	2.50	2.52	2.52	2.50	2.52	2.52	2.56	2.57	2.59
	2	2.49	2.54	2.51	2.52	2.53	2.53	2.55	2.56	2.60

Table 12: Error breakdown on MBPP (share of test items, pooled over seeds).

Condition	Pass	Assertion	TypeError	NameError	SyntaxError	Other runtime	Timeout
Real only	29.1%	55.0%	8.3%	1.8%	1.0%	4.6%	0.2%
1:1	30.8%	53.0%	7.8%	2.2%	1.3%	4.3%	0.5%
2:1	32.9%	54.3%	7.0%	1.6%	1.1%	2.8%	0.4%
3:1	32.6%	54.6%	6.5%	1.7%	1.2%	2.8%	0.6%
5:1	34.1%	52.2%	7.3%	2.0%	0.8%	3.1%	0.4%
10:1	33.7%	53.6%	6.9%	1.0%	1.5%	3.1%	0.2%
20:1	34.1%	52.9%	7.4%	1.3%	1.3%	2.7%	0.4%
30:1	34.6%	51.9%	7.9%	1.7%	1.1%	2.5%	0.3%
Zero-shot student	34.8%	48.0%	7.2%	2.0%	5.0%	2.8%	0.2%

Table 13: Error breakdown on GSM8K (share of test items, pooled over seeds).

Condition	Correct	Wrong answer	No answer line
Real only	32.6%	66.2%	1.2%
1:1	35.5%	62.4%	2.1%
2:1	36.2%	58.3%	5.5%
3:1	36.3%	60.3%	3.4%
5:1	37.3%	59.5%	3.1%
10:1	38.3%	57.9%	3.7%
20:1	38.0%	58.3%	3.7%
30:1	41.3%	56.2%	2.5%
50:1	41.7%	54.7%	3.7%